

人工智能 A 2022-2023 学年期末考试真题

1. (10 points) Rewrite the following rule to make it more efficient

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(defrule bad-rule
  (items (x ?x1)(y ?y1)(z ?z1))
  (items (x ?x2)(y ?y2)(z ?z2))
  (items (x ?x3)(y ?y3)(z ?z3))
  (test (and (or (eq ?x3 green)
                  (eq ?x3 red))
              (eq ?z2 ?y3)
              (> ?y1 ?x1)
              (< ?z1 ?x1)
              (neq ?z3 ?x1))))
```

=>

3. (10 points) Please use Genetic Algorithm to calculate the max value of the function:

$$f(x) = x + 10 \cdot \sin(5 \cdot x) + 7 \cdot \cos(4 \cdot x), x \in [0, 9]$$

Just describe your solution briefly and you don't need to show every step of your solution in details. When necessary, you can write some pseudocodes to make your solution clear.

4. (10 points) Use resolution to solve the following problem.

- (1) Every child loves monkey king.
- (2) Everyone who loves monkey king loves any monkey
- (3) Blythe is a monkey, and Blythe has a red ass.
- (4) Anything which has a red ass is weird or is sick.
- (5) Blythe is not sick.
- (6) Mary does not love anything which is weird

Question: Is Mary a child?

5. (10 points) Given the Rough set information table, where e is decision attribute, calculate its reduction and write the corresponding rules.

U	a	b	c	d	e
(1)	1	0	2	2	1
(2)	0	1	1	1	2
(3)	2	1	1	1	0
(4)	0	0	1	0	0
(5)	1	1	0	2	0
(6)	2	1	1	2	1
(7)	0	2	0	2	2
(8)	1	1	2	2	1

6. (10 points) Fuzzy Set. Given numeric truth values.

$$\mu(A) = 0.95/0.6 + 0.65/0.9 + 0.4/1.0$$

$$\mu(B) = 0.35/0.6 + 0.15/0.9 + 0.7/1.0$$

calculate the fuzzy logic truth of the following:

(1) NOT A

(2) A AND B

(3) A OR B

(4) $A \rightarrow B$

(5) $B \rightarrow A$

7. (15 points) Given a hidden Markov model consisting of a box and a ball, $\lambda = (A, B, \pi)$

$$A = \begin{bmatrix} 0.5 & 0.1 & 0.4 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}, B = \begin{bmatrix} 0.5 & 0.5 \\ 0.4 & 0.6 \\ 0.7 & 0.3 \end{bmatrix}, \pi = (0.5, 0.3, 0.2)^T$$

A is the state-transition probabilities matrix. B is the emission probabilities matrix. π is the initial state probabilities matrix.

Box state collection $\{box\ 1, box\ 2, box\ 3\}$

Color of ball collection $\{red, write\}$

Let $T = 3$, observation sequence $O = (red, white, white)$. Please compute the best path.

8. (10 points) let $p = f(x)$ be the output of the network' s soft-max layer of some neural network classifier with K layers when the network' s input is x . The classifier' s output is then

$$\hat{y} = \arg \max p$$

if y_n is the true label corresponding to training input x_n , the loss is $l_n = l(y_n, f(x_n))$ for some appropriate loss function $l(y, p)$

we saw in class that if $x^{(k)}$ is the output from layer k and $w^{(k)}$ is a vector with all the parameters in layer k , then back-propagation computes the partial derivatives by the following recursion, where $x^{(0)} = x$ is the input to the network and $x^{(K)} = p$

$$\frac{\partial l_n}{\partial w^{(k)}} = \frac{\partial l_n}{\partial x^{(k)}} \frac{\partial x^{(k)}}{\partial w^{(k)}} \text{ for } k = K, \dots, 1$$

$$\frac{\partial l_n}{\partial x^{(k-1)}} = \frac{\partial l_n}{\partial x^{(k)}} \frac{\partial x^{(k)}}{\partial x^{(k-1)}} \text{ for } k = K, \dots, 1$$

$$\frac{\partial l_n}{\partial x^{(K)}} = \frac{\partial l_n}{\partial p}$$

The derivatives above are computed for the n -th training sample (x_n, y_n) and for the value of $w^{(k)}$ that are current at any given point during training.

Suppose that the network has only fully connected layers (with ReLU nonlinearities) before the soft-max.

Refer in detail to the equations given above to explain clearly why training would not work if the parameter vector

$$w = [w^{(1)}, \dots, w^{(k)}]^T$$

is initialized with zeros for training. (Hints: consider $b = 0$ and $b \neq 0$.)

9. (15 points) Consider a convolutional network model A , which contains convolutional layer 1, pooling layer and convolutional layer 2 respectively.

The parameters of each layer are shown in the table: assume that the input image size is $1 \times 32 \times 32$ (including 1 channel, each channel size is 32×32).

	Input Channel	Output Channel	Kernel Size	Step Size	Padding Size
Convolution Layer 1	1	1	3	1	1
Max Pooling			2	2	0
Convolution Layer 2	1	3	3	1	0

- (1) (5 points) If the input

$$I = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix},$$

convolution kernel

$$K = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 2 \\ 3 & 1 & 4 \end{bmatrix},$$

padding value is 0, calculate the output of convolution layer 1

- (2) (5 points) 1×1 convolution is the layer with convolution kernel size = 1, step size = 1, padding size = 0. What is the meaning of 1×1 convolution?